

Harshita Karnam

Engineering Undergraduate

Second-year B.Tech CSE student at Indian Institute of Information Technology Sri City with hands-on experience in full-stack web development and AI/ML research. Built and deployed production-grade web apps; published research (under review) on LLM-guided RL for adaptive traffic control. Strong problem-solving, teamwork, and communication skills.

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GitHub: github.com/harshitak13

Portfolio: portfolio.vercel.app



EDUCATION

B.Tech — Computer Science & Engineering *2024 – 2028 (Completed 2nd Year)*
Indian Institute of Information Technology, Sri City, India

TECHNICAL SKILLS

Languages	Python, C, C++, HTML, CSS, JavaScript, TypeScript
Web / Full-Stack	React, Next.js, Node.js, REST APIs, JWT, MongoDB, Firebase, Vercel
ML / RL	PyTorch, OpenAI Gym, SUMO, DQN, DDQN, PPO, CoLight, Multi-Agent RL
Concepts	Graph Attention Networks, LLM Integration, Reward Shaping, Policy Gradient
Tools	Git / GitHub, Linux, VS Code

RESEARCH

SafeGAT (*Manuscript Under Review*) — LLM-augmented Graph Attention RL framework for adaptive traffic signal control; outperforms pure RL baselines on unseen road topologies.

PROJECTS

Velora Market — Multi-Vendor Marketplace *TypeScript, Next.js, MongoDB, JWT*

- Full-stack marketplace with buyer/seller/admin dashboards, role-based auth, and MongoDB persistence; Next.js API backend with HTTP-only refresh cookies.
- [GitHub: harshitak13/Velora](https://github.com/harshitak13/Velora) [Live: velora-three-irid.vercel.app](https://velora-three-irid.vercel.app)

CinePlex — React Movie Discovery App *React, Firebase, TMDB API*

- Trending film browser with genre search, trailers, personal watchlist, infinite scroll, and actor pages; Firebase Auth & Firestore for user data.
- [GitHub: harshitak13/cineplex](https://github.com/harshitak13/cineplex) [Live: cineplex-aupw.vercel.app](https://cineplex-aupw.vercel.app)

SafeGAT — LLM-Integrated RL Traffic Signal Control *PyTorch, SUMO, LLM*

- Hybrid LLM + GAT-RL architecture for adaptive traffic control; improved generalisation across unseen intersections via LLM-derived auxiliary policy guidance.
- [GitHub: harshitak13/SafeGat](https://github.com/harshitak13/SafeGat)

RL Traffic Signal Control — DQN / DDQN / PPO / CoLight

- Benchmarked four RL algorithms on single and 4×4 multi-intersection grids; DDQN and CoLight led on reward, queue length, and travel time.
- [GitHub: harshitak13/RL-TSC](https://github.com/harshitak13/RL-TSC)

EXPERIENCE

Director, Management Dept. — Entrepreneurship Cell, IIIT Sri City *Sept 2025 – Present*

- Organised AI/ML and developer workshops; led cross-functional student teams and tech events.

Member, Developer Relations — Google Developer Groups (GDG) *Sept 2025 – Present*

- Participated in developer events, study jams, and collaborative coding/ML sessions.